

Exercise: 1) a) $2x^2 + y^2 - 4x + 4y = 0$, origin (1, -2), Transform to parallel axes
Pg-30

$$\begin{aligned} \therefore x &= \alpha + x' \\ &= x' + 1 \end{aligned}$$

$$\therefore y = \beta + y' \\ = y' - 2$$

$$\Rightarrow 2x^2 - 4x + y^2 + 4y = 0$$

$$\Rightarrow 2(x^2 - 2 \cdot 1 \cdot x + 1^2) + (y^2 + 2 \cdot y \cdot 2 + 2^2) - 2 - 4 = 0$$

$$\Rightarrow 2(x-1)^2 + (y+2)^2 - 6 = 0$$

$$\Rightarrow 2(x' + 1 - 1)^2 + (y' - 2 + 2)^2 = 6$$

$$\Rightarrow 2x'^2 + y'^2 = 6$$

Pg-30 1) b) $x^2 - 6x + 2y^2 + 7 = 0$, (3, 1) origin

$$\begin{aligned} \therefore x &= 3 + x' \\ \therefore y &= 1 + y' \end{aligned}$$

$$\Rightarrow (x^2 - 2 \cdot x \cdot 3 + 3^2) + 2y^2 + 7 - 9 = 0$$

$$\Rightarrow (x-3)^2 + 2y^2 - 2 = 0$$

$$\Rightarrow (3 + x' - 3)^2 + 2(1 + y')^2 - 2 = 0$$

$$\Rightarrow x'^2 + 2(1 + 2y' + y'^2) - 2 = 0$$

$$\Rightarrow x'^2 + 2 + 4y' + 2y'^2 - 2 = 0$$

$$\Rightarrow x'^2 + 2y'^2 + 4y' = 0$$

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2) i) $x^2 - y^2 = a^2$, transform axes inclined 45°

$$\begin{aligned} x &= x' \cos \theta - y' \sin \theta = x' \frac{\sqrt{2}}{2} - y' \frac{\sqrt{2}}{2} \\ y &= x' \sin \theta + y' \cos \theta = x' \frac{\sqrt{2}}{2} + y' \frac{\sqrt{2}}{2} \end{aligned}$$

$$\Rightarrow x^2 - y^2 = a^2$$

$$\Rightarrow (x+y)(x-y) = a^2$$

$$\Rightarrow \left(x' \frac{\sqrt{2}}{2} - y' \frac{\sqrt{2}}{2} + x' \frac{\sqrt{2}}{2} + y' \frac{\sqrt{2}}{2} \right) \left(x' \frac{\sqrt{2}}{2} - y' \frac{\sqrt{2}}{2} - x' \frac{\sqrt{2}}{2} - y' \frac{\sqrt{2}}{2} \right) = a^2$$

$$\Rightarrow (2x' \frac{\sqrt{2}}{2}) (-2y' \frac{\sqrt{2}}{2}) = a^2$$

$$\Rightarrow -2x'y' = \frac{a^2}{2}$$

$$\Rightarrow -2x'y' = a^2$$

2) ii) $x^2 - y^2 - 2\sqrt{2}x - 10\sqrt{2}y + 2 = 0$

$$\Rightarrow -2x'y' - 2\sqrt{2} \left\{ \frac{1}{\sqrt{2}} (x' - y') \right\} + 10\sqrt{2} \left\{ \frac{1}{\sqrt{2}} (x' + y') \right\} + 2 = 0$$

$$\Rightarrow -2x'y' - 2x' + 2y' - 10x' - 10y' + 2 = 0$$

$$\Rightarrow (-2x'y' - 12x' - 8y' + 2) = 0$$

$$\Rightarrow x'y' + 6x' + 4y' = 1$$

Pg: 30 | Remove first degree terms |

3) a) $3x^2 - 4y^2 - 6x - 8y - 10 = 0$

$\Rightarrow 3(x^2 - 2x) - 4(y^2 + 2y) - 10 = 0$

$\Rightarrow 3(x^2 - 2 \cdot x \cdot 1 + 1^2) - 4(y^2 + 2 \cdot y \cdot 1 + 1^2) - 10 - 3 + 4 = 0$

$\Rightarrow 3(x-1)^2 - 4(y+1)^2 - 9 = 0$

Let,

$x' = x - 1 \quad y' = y + 1$
 $\Rightarrow x = x' + 1 \quad y = y' - 1$

Origin $(-1, +1)$

$3x'^2 - 4y'^2 = 9$

3) b) $2x^2 + 5y^2 - 12x + 10y - 7 = 0$

$\Rightarrow 2(x^2 - 6x) + 5(y^2 + 2y) - 7 = 0$

$\Rightarrow 2(x^2 - 2 \cdot x \cdot 3 + 3^2) + 5(y^2 + 2 \cdot y \cdot 1 + 1^2) - 7 - 18 + 5 = 0$

$\Rightarrow 2(x-3)^2 + 5(y+1)^2 - 30 = 0$

Let,

$x' = x - 3 \quad y' = y + 1$
 $\Rightarrow x = x' + 3 \quad y = y' - 1$

Origin $(3, -1)$

$2x'^2 + 5y'^2 = 30$

3) c) $3x^2 - 4y^2 + 6x + 24y - 135 = 0$

$\Rightarrow 3(x^2 + 2 \cdot x \cdot 1 + 1^2) - 4(y^2 - 2 \cdot y \cdot 3 + 3^2) - 135 - 3 + 36 = 0$

$\Rightarrow 3(x+1)^2 - 4(y-3)^2 - 102 = 0$

Let,

$x' = x + 1 \quad y' = y - 3$
 $x = x' - 1 \quad y = y' + 3$

Origin $(-1, +3)$

$3x'^2 - 4y'^2 = 0$

5) Transform eqn $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$
through point $(2, -1)$, inclined at angle $(-\tan^{-1} \frac{4}{3})$

$$\therefore 11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0 \quad \left| \begin{array}{l} x = x' + \alpha \\ y = y' + \beta \end{array} \right. \quad \begin{array}{l} (2, -1) \\ \alpha \beta \end{array}$$

$$\Rightarrow 11(x' + 2)^2 + 4(y' + (-1))^2 + 24(x' + 2)(y' - 1) - 20(x' + 2) - 40(y' - 1) - 5 = 0$$

$$\Rightarrow 11(x'^2 + 4x' + 4) + 4(y'^2 - 2y' + 1) + 24(x'y' - x' + 2y' - 2) - 20x' - 40 - 40y' + 40 - 5 = 0$$

$$\Rightarrow 11x'^2 + 44x' + 44 + 4y'^2 - 8y' + 4 + 24x'y' - 24x' + 48y' - 48 - 20x' - 40y' - 5 = 0$$

$$\Rightarrow 11x'^2 + 4y'^2 + 24x'y' = 5$$

$$\Rightarrow 11x'^2 + 4y'^2 + 24x'y' = 5 \rightarrow \text{transformed through point } (2, -1)$$

$$\begin{array}{l} x = x' \cos \theta - y' \sin \theta \\ y = x' \sin \theta + y' \cos \theta \end{array} \quad \left| \begin{array}{l} \cos(-\tan^{-1} \frac{4}{3}) = \frac{3}{5} \\ \sin(-\tan^{-1} \frac{4}{3}) = -\frac{4}{5} \end{array} \right. \quad \left| \begin{array}{l} x = x' \frac{3}{5} + y' \frac{4}{5} \\ y = x' \frac{4}{5} + y' \frac{3}{5} \end{array} \right.$$

$$\Rightarrow 11 \left(x' \frac{3}{5} + y' \frac{4}{5} \right)^2 + 4 \left(x' \frac{4}{5} + y' \frac{3}{5} \right)^2 + 24 \left(x' \frac{3}{5} + y' \frac{4}{5} \right) \left(x' \frac{4}{5} + y' \frac{3}{5} \right) = 5$$

$$\Rightarrow 11 \left(\frac{3x' + 4y'}{5} \right)^2 + 4 \left(\frac{-4x' + 3y'}{5} \right)^2 + 24 \left(\frac{3x' + 4y'}{5} \right) \left(\frac{-4x' + 3y'}{5} \right) = 5 \times 25$$

multiply all by 25

$$\Rightarrow 11(9x'^2 + 24x'y' + 16y'^2) + 4(16x'^2 - 24x'y' + 9y'^2) + 24(-12x'^2 + 9x'y' - 16x'y' + 12y'^2) = 125$$

$$\Rightarrow 99x'^2 + 264x'y' + 176y'^2 + 64x'^2 - 96x'y' + 36y'^2 - 288x'^2 + 168x'y' - 288y'^2 = 125$$

$$\Rightarrow \frac{-125x'^2 + 500y'^2}{125} = \frac{125}{125} \quad \left(\text{Divide by } 125 \text{ or } -125 \right)$$

$$\Rightarrow -x'^2 + 4y'^2 = 1$$

$$\Rightarrow \cancel{(x'^2 - 4y'^2)} = 1$$

$$\Rightarrow x'^2 - 4y'^2 + 1 = 0$$

6) Determine the angle through which axes must be rotated to remove xy term in equation:

$$7x^2 - 6\sqrt{3}xy + 13y^2 = 16$$

Ans: $ax^2 + 2hxy + by^2 = 0$

$$\theta = \frac{1}{2} \tan^{-1} \frac{2h}{a-b}$$

$$\theta = \frac{1}{2} \tan^{-1} \frac{2 \cdot 3\sqrt{3}}{7-13}$$

$$\theta = -30^\circ$$

$$x^2 \cos^2 \theta - y^2 \sin^2 \theta$$

$$= x^2 \frac{\sqrt{3}}{2} - y^2 \frac{1}{2}$$

$$y^2 \sin^2 \theta + x^2 \cos^2 \theta$$

$$= x^2 \frac{1}{2} + y^2 \frac{\sqrt{3}}{2}$$

Find inclination of axes to angle (-30°)

$$7 \left(x \frac{\sqrt{3}}{2} - y \frac{1}{2} \right)^2 - 6\sqrt{3} \left(x \frac{\sqrt{3}}{2} - y \frac{1}{2} \right) \left(x \frac{1}{2} + y \frac{\sqrt{3}}{2} \right) + 13 \left(x \frac{1}{2} + y \frac{\sqrt{3}}{2} \right)^2 = 16$$

$$7 \left(\frac{x\sqrt{3}-y}{2} \right)^2 - 6\sqrt{3} \left(\frac{x\sqrt{3}-y}{2} \right) \left(\frac{x+\sqrt{3}y}{2} \right) + 13 \left(\frac{x+\sqrt{3}y}{2} \right)^2 = 16 \times 4$$

multiply all by 4

$$7(x\sqrt{3}-y)^2 - 6\sqrt{3}(x\sqrt{3}-y)(x+\sqrt{3}y) + 13(x+\sqrt{3}y)^2 = 64$$

$$7(3x^2 - 2\sqrt{3}xy + y^2) - 6\sqrt{3}(x^2\sqrt{3} + 3xy - xy - y^2\sqrt{3}) + 13(x^2 + 2\sqrt{3}xy + 3y^2) = 64$$

$$7(3x^2 - 2\sqrt{3}xy + y^2) - 6\sqrt{3}(x^2\sqrt{3} + 2xy - y^2\sqrt{3}) + 13(x^2 + 2\sqrt{3}xy + 3y^2) = 64$$

$$21x^2 - 14\sqrt{3}xy + 7y^2 - 18x^2 - 12\sqrt{3}xy + 18y^2 + 13x^2 + 26\sqrt{3}xy + 39y^2 = 64$$

$$16x^2 + 64y^2 = 64$$

$$x^2 + 4y^2 = 4$$

7) point (h, k) , transforming to parallel, prove that

eqn $2x^2 + y^2 - xy - 5x - 4y + 11 = 0$, can spot...

Ans: $x = x' + \alpha$ (h, k) $x = x' + h$
 $y = y' + \beta$ α, β $y = y' + k$

$$2(x' + h)^2 + (y' + k)^2 - (x' + h)(y' + k) - 5(x' + h) - 4(y' + k) + 11 = 0$$

$$2(x'^2 + 2x'h + h^2) + y'^2 + 2y'k + k^2 - x'y' - x'k - y'h - hk - 5x' - 5h - 4y' - 4k + 11 = 0$$

$$2x'^2 + 4x'h + h^2 + y'^2 + 2y'k + k^2 - x'y' - x'k - y'h - hk - 5x' - 5h - 4y' - 4k + 11 = 0$$

1st & 2nd degree,
 x common terms,
 y common terms,
 constant

$[x^2, y^2 = 2^{nd} \text{ degree}]$

$$2x'^2 - x'y' + y'^2 + x'(4h - k - 5) + y'(2k - h - 4) + h^2 + k^2 - hk - 5h - 4k + 11 = 0$$

Since, question

question ask, to keep 2nd degree terms:

$$2x^2 - xy + y^2 = 0$$

question ask, to eliminate 1st degree terms:

$$4h - k - 5 = 0 \rightarrow (i)$$

$$2k - h - 4 = 0 \rightarrow (ii)$$

$$\rightarrow (i) \times (ii) + (i)$$

$$\rightarrow -2k + 8h = 10$$

$$2k - h = 4$$

$$7h = 14$$

$$h = 2$$

$$8 - k - 5 = 0$$

$$k = 3$$

(10) Transform axes inclined at 30° to original, find

$$eq^n = x^2 + 2\sqrt{3}xy - y^2 = 2a^2$$

$$\begin{aligned} x &= x' \cos \theta - y' \sin \theta \\ y &= x' \sin \theta + y' \cos \theta \end{aligned}$$

$$\begin{aligned} &= (x' \cos \theta - y' \sin \theta)^2 + 2\sqrt{3}(x' \sin \theta + y' \cos \theta)(x' \cos \theta - y' \sin \theta) - (x' \sin \theta + y' \cos \theta)^2 = 2a^2 \\ &= x'^2 \cos^2 \theta - 2x'y' \sin \theta \cos \theta + y'^2 \sin^2 \theta + 2\sqrt{3}x'x' \sin \theta \cos \theta - 2\sqrt{3}x'y' \sin^2 \theta - 2\sqrt{3}y'y' \cos^2 \theta + y'^2 \cos^2 \theta = 2a^2 \\ &= x'^2 (\cos^2 \theta + \sin^2 \theta) + y'^2 (\sin^2 \theta + \cos^2 \theta) + 2x'y' (\sqrt{3} \sin \theta \cos \theta - \sin^2 \theta - \cos^2 \theta) = 2a^2 \\ &= x'^2 + y'^2 + 2x'y' (\sqrt{3} \sin \theta \cos \theta - \sin^2 \theta - \cos^2 \theta) = 2a^2 \end{aligned}$$

$$\Rightarrow 3x'^2 - 2\sqrt{3}x'y' + y'^2 + 2\sqrt{3}(\sqrt{3}x'^2 + 3xy - xy\sqrt{3}y^2) - x^2 - 2\sqrt{3}xy - 3y^2 = 8a^2$$

$$\Rightarrow 3x'^2 - 2\sqrt{3}x'y' + y'^2 + 6x'^2 + 4\sqrt{3}xy - 6y^2 - x^2 - 2\sqrt{3}xy - 3y^2 = 8a^2$$

$$\Rightarrow 8x^2 - 8y^2 = 8a^2$$

$$\Rightarrow x^2 - y^2 = a^2$$

$$\boxed{4, 8 \text{ अंश}}$$

$$\textcircled{1} + \textcircled{2} = 0 = 2 - 4 - 11$$

$$\textcircled{1} - \textcircled{2} = 0 = 11 - 4 - 11$$

$$\textcircled{1} + \textcircled{2} \times \textcircled{3}$$

$$0 \textcircled{1} - 11 \textcircled{2} + 11 \textcircled{3} =$$

$$11 = 11 - 11$$

$$11 = 11$$

$$\boxed{8 = 11}$$

$30[5, 6, 7, 9], 31 [\text{except 8 all}]$
 $Pg-24[4]$
 $-28[9]$

CGTava
change of axes

8/8/2

$$ax^2 + 2hxy + by^2 = 0$$

$$\theta = \frac{1}{2} \tan^{-1} \frac{2h}{a-b}$$

(Two points given, value of θ to get from origin point to other point)

$\Rightarrow$ The equation $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ is transformed to $4x'^2 + 2y'^2 = 1$ when referred to rectangular axes through the point $(2, 3)$. Find the inclination of the latter axes to the former.

$$\begin{aligned}
 x &= x' + 2 \\
 y &= y' + 3
 \end{aligned}$$

$$\therefore \Rightarrow 3(x'+2)^2 + 2(y'+3)^2 + 2(x'+2)(y'+3) - 18(x'+2) - 22(y'+3) + 50 = 0$$

$$\Rightarrow 3(x'^2 + 4x' + 4 + y'^2 + 6y' + 9) + 2(x'y' + 3x' + 2y' + 6) - 18x' - 36 - 22y' - 66 + 50 = 0$$

$$\Rightarrow \underline{3x'^2} + \underline{12x'} + \underline{39} + \underline{3y'^2} + \underline{18y'} + \underline{2x'y'} + \underline{6x'} + \underline{4y'} + 12 - 18x' - 22y' + 50 = 0$$

$$\Rightarrow 3x'^2 + 3y'^2 + 18x' + 22y' + 2x'y' - 18x' - 22y' + 40 = 0$$

$$\Rightarrow 3x'^2 + 3y'^2 + 2x'y' = 0$$

$$\Rightarrow 3x'^2 + 3y'^2 + 2x'y' = 1$$

$$\Rightarrow 3x'^2 + 2 \cdot 1 x'y' + 3y'^2 = 1$$

$$\therefore \theta = \frac{1}{2} \cdot \tan^{-1} \frac{2 \cdot 1}{3-3}$$

$$\theta = \frac{1}{2} \cdot \tan^{-1} \infty$$

$$\Rightarrow \theta = \frac{1}{2} \cdot 90^\circ$$

$$\boxed{\theta = 45^\circ}$$

12/8/18

$$x = x' \cos \theta - y' \sin \theta$$

$$y = x' \sin \theta + y' \cos \theta$$

$$x = x' \frac{\sqrt{2}}{2} - y' \frac{\sqrt{2}}{2}$$

$$y = x' \frac{\sqrt{2}}{2} + y' \frac{\sqrt{2}}{2}$$

$$\Rightarrow 3x^2 + 3y^2 + 2xy = 1$$

$$\Rightarrow 3 \left(x' \frac{\sqrt{2}}{2} - y' \frac{\sqrt{2}}{2} \right)^2 + \left(x' \frac{\sqrt{2}}{2} + y' \frac{\sqrt{2}}{2} \right)^2 + 2 \left(x' \frac{\sqrt{2}}{2} - y' \frac{\sqrt{2}}{2} \right) \left(x' \frac{\sqrt{2}}{2} + y' \frac{\sqrt{2}}{2} \right)$$

$$\Rightarrow 3 \left\{ 2 \left(\frac{1}{2} x'^2 \right) + 2 \left(\frac{1}{2} y'^2 \right) + 2 \left(x' \frac{1}{2} - y' \frac{1}{2} \right) \right\} = 1$$

$$\Rightarrow 3x'^2 + 3y'^2 + x'^2 - y'^2 = 1$$

$$\boxed{\Rightarrow 4x'^2 + 2y'^2 = 1}$$

$$\begin{aligned} x' + y' &= x \\ x' - y' &= y \end{aligned}$$

$$\begin{aligned} 2 \left\{ x'^2 \frac{1}{2} - y'^2 \frac{1}{2} \right\} \\ 2 \left(\frac{x'^2 - y'^2}{2} \right) \end{aligned}$$

$$\boxed{2x'^2 + 2y'^2 + x'^2 - y'^2 = 1}$$

$$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{16}$$